Chapter test

20 Ecosystems

Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

Score: / 40

A. Concept checking (10 marks, 1 mark each)

1. In ecology, a population refers to

   A. a region with a distinctive climate and community of organisms.
   B. organisms living together and interacting in the same habitat.
   C. organisms of a species living together in the same habitat.
   D. a group of organisms that can interbreed. ☐

2. Which of the following is / are recycled in an ecosystem?

   (1) water
   (2) nitrogen
   (3) energy

   A. (2) only
   B. (1) and (2) only
   C. (2) and (3) only
   D. (1), (2) and (3) ☐

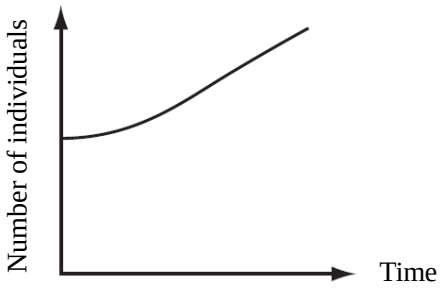
3. A lichen is a close association between a fungus and an alga, which cannot survive independently. This is an example of

   A. predation.
   B. commensalism.
   C. mutualism.
   D. parasitism. ☐

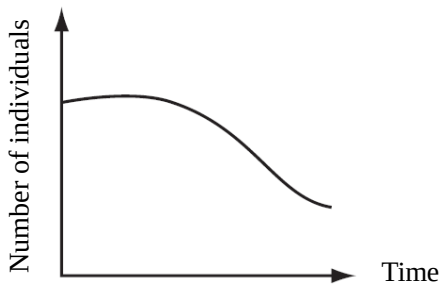
© 2019 Aristo Educational Press Ltd.

★ 4. Species P feeds on species Q. Which of the following graphs correctly predicts the change in the number of species Q when species P is removed from the habitat?

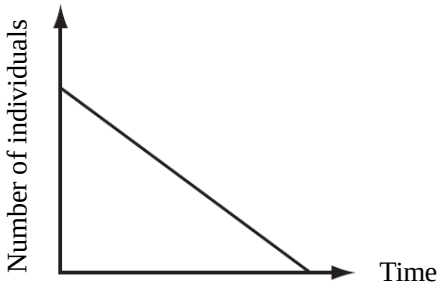
A.



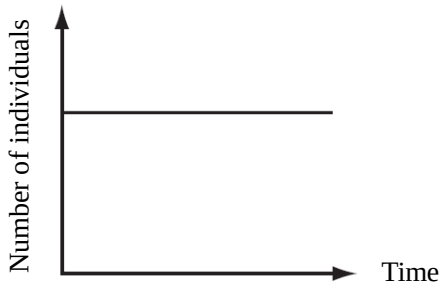
B.



C.



D.



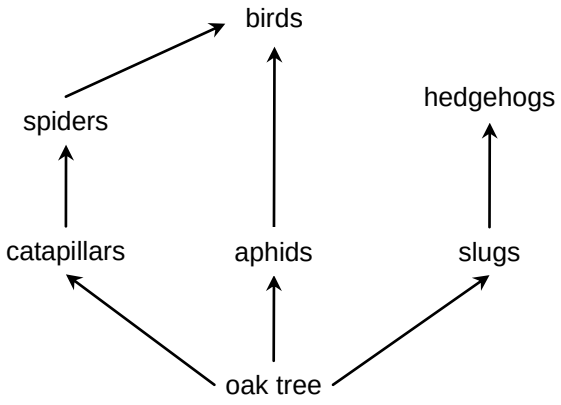
( )

5. Which of the following is ***not*** a producer?

A. cyanobacteria
B. *Euglena*
C. bread mold
D. moss

( )

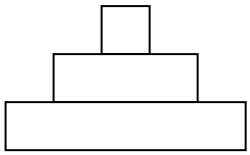
Directions: Questions 6 and 7 refer to the diagram below, which shows part of a food web in an oak woodland.



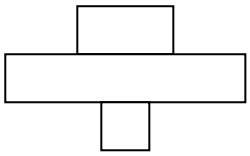
© 2019 Aristo Educational Press Ltd.

6. Which of the following diagrams represents the pyramid of numbers for the food chain consisting of the oak tree, aphids and birds?

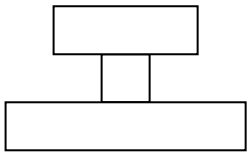
A.



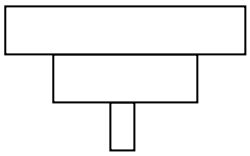
B.



C.



D.



☐

★ 7. The use of insecticides has killed most of the caterpillars and aphids. Which of the following shows the effects on the numbers of slugs and birds?

	Number of slugs	***Number of birds***
A.	remains the same	decreases
B.	remains the same	remains the same
C.	increases	decreases
D.	increases	remains the same

☐

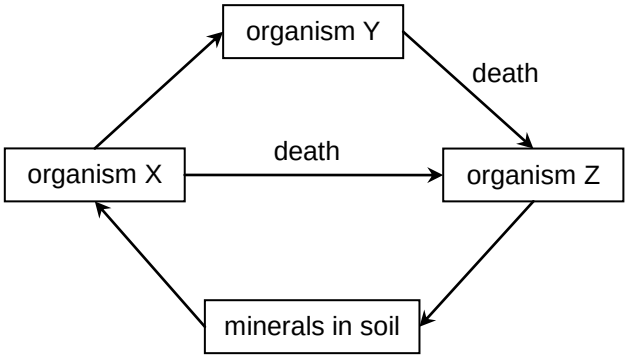
8. Which of the following food chains would pass on the highest proportion of energy from the Sun to humans?

A. beans → humans
B. corn → pigs → humans
C. grass →cattle → humans
D. algae → insects → fish → humans

☐

© 2019 Aristo Educational Press Ltd.

9. The diagram below shows the flow of nutrients in an ecosystem.



Which of the following correctly identifies the three groups of organisms, X, Y and Z?

	X	***Y***	***Z***
A.	consumers	decomposers	producers
B.	decomposers	consumers	producers
C.	producers	decomposers	consumers
D.	producers	consumers	decomposers

10. The photograph below shows the roots of a leguminous plant.



What chemical changes is carried out by the bacteria inside swelling X?

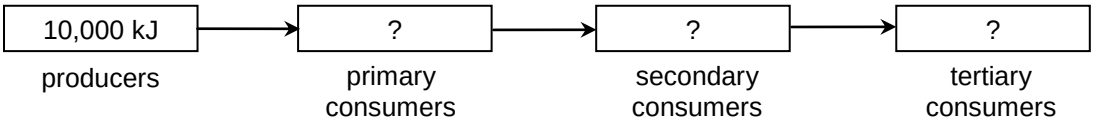
A. ammonium compounds → nitrates
B. nitrates → nitrogen gas
C. nitrogen gas → nitrates
D. nitrogen gas → ammonium compounds

Question	1	2	3	4	5	6	7	8	9	10
Related section	20.1	20.1	20.2	20.2	20.2	20.3	20.2	20.3	20.6	20.5

© 2019 Aristo Educational Press Ltd.

B. Problem solving (20 marks)

★ 11. The diagram below shows the energy flow through a food chain in a terrestrial habitat. The total energy in the tissues of the producers is shown in the box.



(a) In what form does energy enter the food chain? (1 mark)

(b) In what form is energy stored in the producers? (1 mark)

(c) Suppose only 10% of the energy in any trophic level is transferred to the next trophic level, calculate the total energy in the tertiary consumers. (2 marks)

(d) Explain why not all of the energy in one trophic level is transferred to the next trophic level. (3 marks)

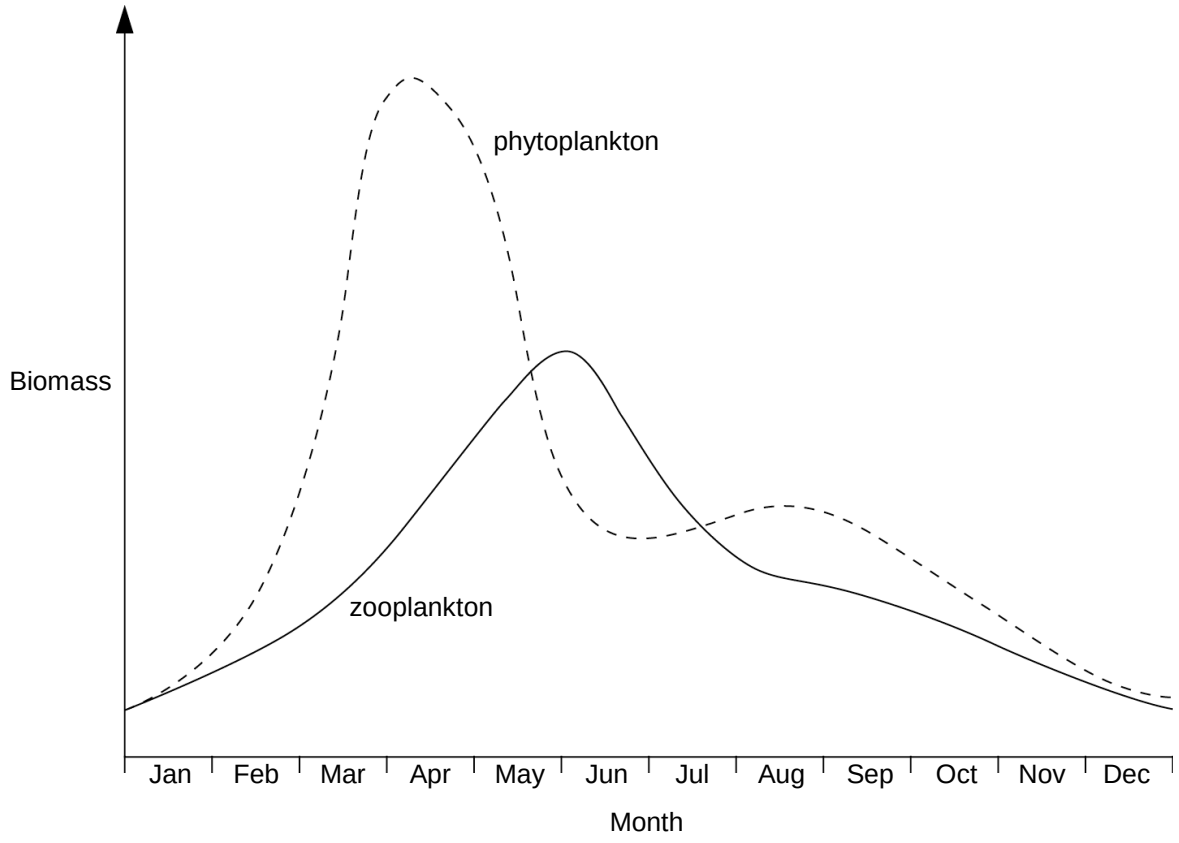
(e) If all the secondary consumers were removed from the habitat and the tertiary consumers could feed on the primary consumers, suggest how the number of the tertiary consumers would change. Explain your answer. (3 marks)

© 2019 Aristo Educational Press Ltd.

(f) Energy also flows from all organisms to a group of organisms not shown in the diagram. Name this group of organisms and explain its importance. (4 marks)

(Total: 14 marks)

★ 12. In an investigation, samples of water were collected from a lake at intervals over a year and the biomasses of phytoplankton and zooplankton were determined. The results are shown in the graph below.



© 2019 Aristo Educational Press Ltd.

(a) In the space provided below, draw the pyramid of biomass for the phytoplankton and zooplankton in June. (2 marks)

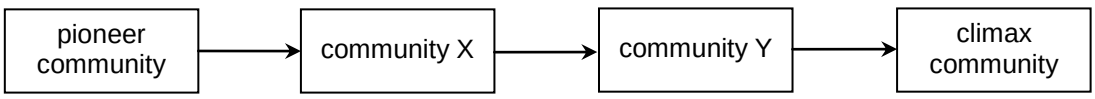
(b) Explain the shape of the pyramid of biomass you have drawn. (2 marks)

(c) In winter, the period of daylight becomes shorter. Explain how this affects the phytoplankton population (2 marks)

(Total: 6 marks)

C. Exam challenge (10 marks)

★★ 13. A field previously used for growing crops was abandoned and developed into woodland. The diagram below shows the succession of communities in the field.



© 2019 Aristo Educational Press Ltd.

(a) Identify, with a reason, the type of succession that occurred in the field. (2 marks)

(b) Suggest a modification that community X may make to its habitat which allows colonization by community Y. (1 mark)

(c) Compare the abundance and diversity of organisms in the pioneer community and the climax community. Explain the difference. (3 marks)

(d) Describe how you would use the mark-and-recapture method to estimate the number of small animals in an area of woodland. (4 marks)

(Total: 10 marks)

© 2019 Aristo Educational Press Ltd.

Marking scheme

A. Concept checking (10 marks, 1 mark each)

1.	C	2.	B	3.	C	4.	A	5.	C
6.	B	7.	C	8.	A	9.	D	10.	D

(1 mark each)

B. Problem solving (20 marks)

11. (a) Solar energy / Light energy (1)

(b) Chemical energy in organic compounds / starch (1)

(c) 10,000 kJ × 10% × 10% × 10% (1)

= 10 kJ (1)

(d) The energy stored in the uneaten body parts is not passed to the next trophic level. (1)

Energy is lost / trapped in the undigested / unabsorbed food. (1)

Some energy is lost by the organisms at the higher trophic level through excretion / respiration / in the form of heat. (1)

(e) The number of the tertiary consumers would increase. (1)

With one less trophic level, (1)

less energy is lost / more energy is conversed, (1)

so that a larger number of tertiary consumers can be sustained.

(f) Decomposers (1)

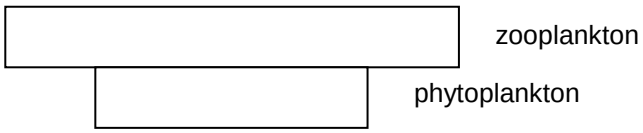
They break down dead organic matter (1)

into inorganic compounds which can be absorbed by producers (1)

so that materials are recycled in the ecosystem. (1)

(Total: 14 marks)

12. (a)



Correct shape (1)

Correct labels (1)

(b) The phytoplankton reproduces very rapidly (1)

so that its small biomass can support the larger biomass of the zooplankton. (1)

(c) The phytoplankton cannot obtain sufficient light to carry out photosynthesis. (1)

Thus, the growth rate becomes slower and the population decreases. (1)

(Total: 6 marks)

© 2019 Aristo Educational Press Ltd.

C. Exam challenge (10 marks)

13. (a) Secondary succession (1)
Crops (and other plants) were present before. (1)

(b) It makes the soil more fertile / increases the nutrient content of the soil. /
It makes the soil thicker. /
The plant roots stabilize the soil.
(Accept other reasonable answers.) (1)

(c) The climax community has a greater abundance and diversity of organisms than the pioneer community. (1)
This is because there are more producers, providing more food to consumers. (1)
The more complex ecosystem provides a greater variety of habitats so that more species can live there. (1)

(d) Capture a sample of animals in the area, and count the number (N_1). (1)
Mark the animals in a harmless way and then release them. (1)
After some time, capture a second sample in the same area, and count the total number (N_2) and the number of marked individuals (N). (1)
Number of animals in the population = $N_1 \times N_2 / N$ (1)

(Total: 10 marks)

© 2019 Aristo Educational Press Ltd.